

MADIGA MALLIKARJUNA

Java Full-Stack Developer | IoT Systems Architect

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PROFESSIONAL SUMMARY

Performance-driven Senior Full-Stack Engineer and IoT Solutions Architect with 3+ years of proven experience in building and deploying mission-critical Smart City systems, Integrated Command & Control Centers (ICCC), and high-throughput IoT production networks. Expert in Java, Spring Boot, Angular, and real-time streaming pipelines. Highly skilled in designing secure, resilient API gateways, edge data gateways, custom authentication layers, telemetry pipelines, and high-availability Linux infrastructures.

CORE SKILLS & TECHNOLOGIES

Languages & Frameworks: Java, TypeScript, Spring Boot, Angular, JavaScript, HTML5/CSS3, Bash Scripting
APIs & Middleware: REST APIs, WebSockets, gRPC, OAuth 2.0 & JWT, Hibernate, JPA, Apache NiFi
Databases & Storage: PostgreSQL, MySQL, Redis, PouchDB, CouchDB, MongoDB, H2 Database
IoT & Protocols: Modbus TCP, OPC UA, BACnet, MQTT, Kafka Streams, Node-RED, Telemetry Gateways
DevOps & Production: Docker, Jenkins CI/CD, SonarQube, Linux System Admin, Prometheus, Grafana

PROFESSIONAL EXPERIENCE

Trinity Mobility Private Ltd - Full Stack Developer | Bengaluru, India (Mar 2022 - Apr 2025)

Role Description: Spearheaded highly available, secure smart city production software and IoT operations.

Notify Service - High-Availability Notification Gateway

- Scaled a performance-driven notification engine supporting 100,000+ users for mission-critical SMS and email alerts.
- Executed custom authentication filters and request-logging mechanisms, managing 1M+ monthly operations with 99.99% reliability.
- Mitigated system downtime by 80% through high-availability response frameworks and Linux-based automated failovers.

Enterprise API Gateway

- Formulated a secure Spring Boot API Gateway serving 12+ Smart City vendors with robust OAuth 2.0 and JWT authentication.
- Architected API solutions with OAuth 2.0 and rate limiting, achieving a 45% reduction in vulnerabilities through robust security.
- Streamlined vendor integration by standardizing API interactions, reducing integration time by 60% for traffic and safety vendors.
- Standardized API communication protocols, which minimized cross-team integration friction by 50%.

Industrial IoT Data Gateway

- Developed a Node-RED gateway decreasing latency by 30%, enabling real-time synchronization between sensors and cloud backends.
- Integrated Modbus TCP, OPC UA, and BACnet protocols, enabling bidirectional communication for 10,000+ edge devices.
- Programmed custom JavaScript logic within Node-RED to perform edge-level data filtering, reducing cloud storage costs by 20%.

R&D: Digital Twinning & Real-Time Analytics

- Pioneered a product design that boosted overall system performance by 20% and decreased operational costs by 15%.
- Improved resource allocation accuracy by 25% via a real-time tracking dashboard, ensuring zero downtime during incidents.
- Achieved a 25% reduction in emergency response times by automating dispatch workflows and live sensor data streaming.

PRODUCTION & LARGE-SCALE IOT OPERATIONS HIGHLIGHTS

- Actively managed live production deployments on multi-node Linux clusters for Smart City systems.
- Standardized Docker container configurations and Jenkins pipelines to guarantee seamless rollouts.
- Executed load-testing mimicking 100K+ concurrent signals to ensure gateway bounds and stability.
- Configured Grafana and Prometheus monitoring metrics to proactively detect memory and socket choke.

EDUCATION & TRAINING

Bachelor of Technology in Electronics and Communication Engineering (ECE)

Jawaharlal Nehru Technological University (JNTU), Anantapur | Andhra Pradesh, India

Professional Course: Full Stack Development Certification Course - JSpiders, Bengaluru

Core Lab Training: RTOS, Microcontrollers, telecommunication networks, computer networks, Embedded Systems